AGA KHAN UNIVERSITY EXAMINATION BOARD

SECONDARY SCHOOL CERTIFICATE

CLASS IX EXAMINATION

MAY 2012

General Mathematics Paper II

Time allowed: 2 hours 20 minutes Marks 45

INSTRUCTIONS

Please read the following instructions carefully.

1. Check your name and school information. Sign that it is correct.

I agree that this is my name and school. **Candidate's signature**

2. RUBRIC. There are TEN questions. Answer ALL TEN questions. Choices are specified inside the paper.

3. When answering the questions:

   Read each question carefully.
   Use a black pencil for diagrams. DO NOT use coloured pencils.
   DO NOT use staples, paper clips, glue or correcting fluid.
   Complete your answer in the allocated space only. DO NOT write outside the answer box.

4. The marks for the questions are shown in brackets ( ).

5. You may use a simple calculator if you wish.

Q.1. (Total 3 Marks)

A transport company has 375 buses in its fleet. 45 buses are not operational due to repair work. What percentage of the buses is operational?



(ATTEMPT EITHER PART a OR PART b OF Q.2.)

Q.2. (Total 4 Marks)

a. Mr. Sajjad has his annual savings of Rs. 350,000, gold of worth Rs. 300,000 and a flat (not in use) of worth Rs. 5,500,000. Find the amount of Zakat payable by him.

b. Mr. Ashraf left behind a property of worth Rs. 187,500. The property was distributed among his heirs; the widow, father and a daughter. The widow gets $\frac{1}{8}$, father gets $\frac{1}{6}$ and daughter gets $\frac{1}{3}$ of the property whereas the remaining property was given to a hospital as charity. Find the share of the daughter and the amount given to the hospital.

(ATTEMPT EITHER PART a OR PART b OF Q.3.)

Q.3. (Total 5 Marks)

a.

i. An electrical appliance dealer buys a washing machine for Rs. 4500.
Find the selling price and profit when profit made is at a rate of 12.5 %. (3 Marks)



ii. Define cost price and selling price. (2 Marks)

(ATTEMPT EITHER PART a OR PART b OF Q.3.)

b.

i. Four partners *A, B, C* and *D* enter in a business with investments of Rs. 450,000, 600,000, 350,000 and 550,000 respectively. They earn profit of Rs. 78,000. Find the share of *D* if each partner gets profit in the proportion to the amount they invested. (3 Marks)

ii. A calculator has a marked price of Rs. 990 on which a discount of 8 % is offered. Find the discounted price. (2 Marks)

Q.4. (Total 3 Marks)

Find the interest rate (in percent) if the principal amount of Rs. 75,000 is invested for a period of 36 months to get a simple interest of Rs. 16,875.

Q.5. (Total 5 Marks)

i. The given table shows the income tax rates for different ranges of income.

Annual Income (in Rs)	Tax Rate
$400{,}000 \leq x \leq 450{,}000$	2.5 %
$450{,}000 \leq x \leq 550{,}000$	3.5 %
$550{,}000 \leq x \leq 650{,}000$	4.5 %
$650{,}000 \leq x \leq 750{,}000$	6.0 %

Find the income tax payable by a person who earns Rs. 35,000 per month from one source and Rs. 26,000 per month from another source. (3 Marks)

ii. Uzma bought a laptop from Dubai worth Rs. 30,000 and paid customs duty of Rs. 5,250. What is the rate of customs duty paid on the price of the laptop? (2 Marks)

(ATTEMPT EITHER PART a OR PART b OF Q.6.)

Q.6. (Total 5 Marks)

a.

i. Find the value of a if $\log_a 343 = 3$. (3 Marks)

ii. Simplify $\frac{0.000724}{3.62}$. Give your answer in scientific notation. (2 Marks)

b. Find the value of k when $\frac{2^0 + 2^2 + 5^{-5} \times 5^6}{2k^2} = 5$. (5 Marks)

Q.7. (Total 4 Marks)

Find TWO arithmetic means between – 1 and 1.

(ATTEMPT EITHER PART a OR PART b OF Q.8.)

Q.8. (Total 5 Marks)

a. If $A = \{1, 2, 3, 4, 5\}$ and $B = \{3, 4, 5, 6, 7\}$, then find $A \Delta B$, where $A \Delta B = (A \cup B) - (A \cap B)$. Also shade the region which represents $A \Delta B$ with the help of a Venn diagram.

b. If $A = \{2, 4, 5\}$ and $B = \{x, y\}$, then

i. find $A \times B$. Also write its domain. (3 Marks)

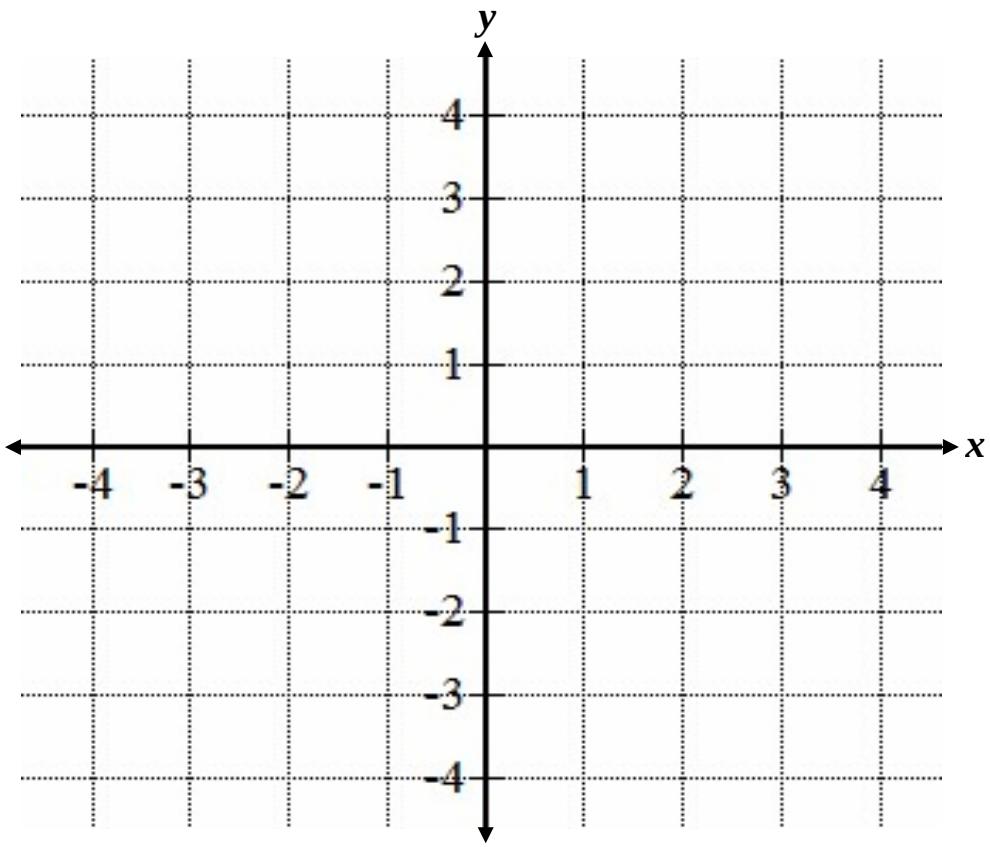
ii. write ONE binary relation from A to B containing three elements. Also write its range. (2 Marks)

Q.9. (Total 5 Marks)

Complete the following table for the equation $2x + 3y = 6$. Also draw its graph on the given graph paper.

x	0	
y		0

Does this equation pass through the third quadrant? Justify your answer.



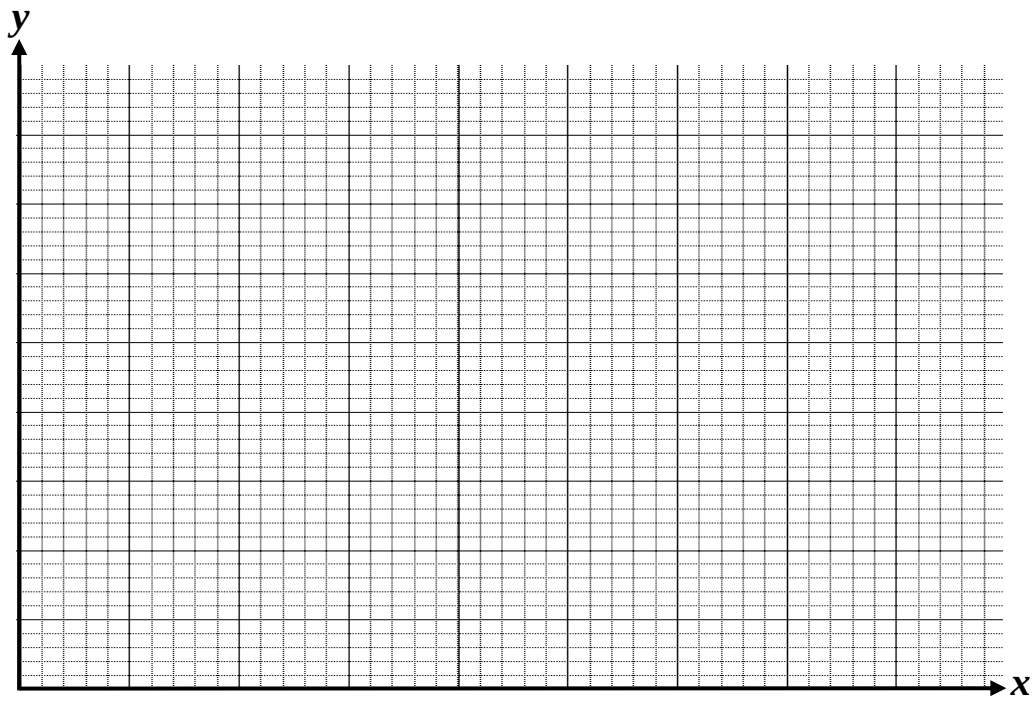
(ATTEMPT EITHER PART a OR PART b OF Q.10.)

Q.10. (Total 6 Marks)

a.

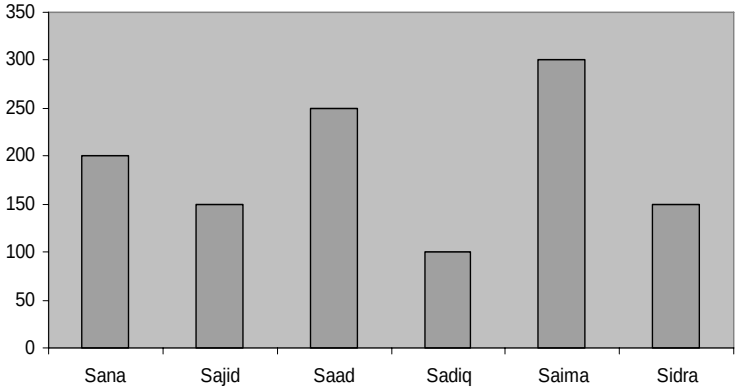
i. Complete the following table and draw a cumulative frequency polygon for the height of 80 plants. (4 Marks)

Height (cm)	Number of plants		
5 – 10	5		
11 – 16	15		
17 – 22	25		
23 – 28	20		
29 – 34	15		



ii. Six friends Sana, Sajid, Saad, Sadiq, Saima and Sidra are fond of collecting stamps. The information about the number of stamps each friend possesses is represented by the following bar chart. (2 Marks)

Find the median of the data.



(ATTEMPT EITHER PART a OR PART b OF Q.10.)

b. The monthly salaries (x) in thousands of 10 factory workers are given as follows. (6 Marks)

15 10 12 11 12 14 10 13 14 16

Find range and standard deviation.



x										
x^2										

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